Cualitativa

color
tacto
rugoso
sabor

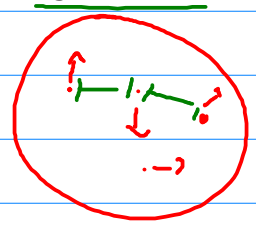
Cuantitativa

masa
volumen
densidad

estados de agregacion

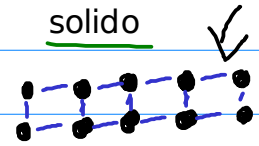
estados de la materia

gaseoso



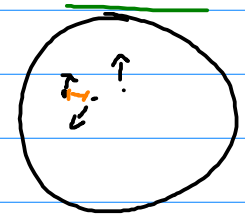
Mas desordenado

solido



El más ordenado

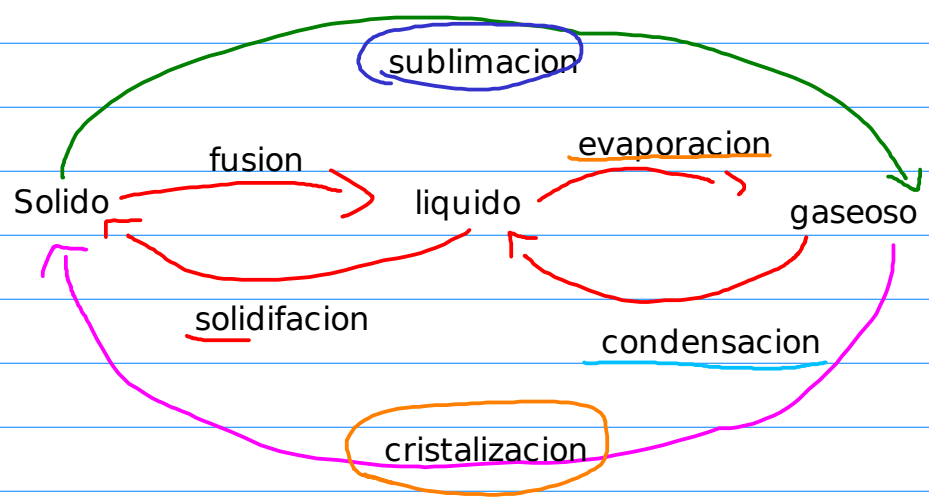
liquido



Mas ordenado
que el gas
pero no tanto

entropia == ordenado

Mas ordenado == menos entropia



calor ----> temperaturas

Variables termodinamicas

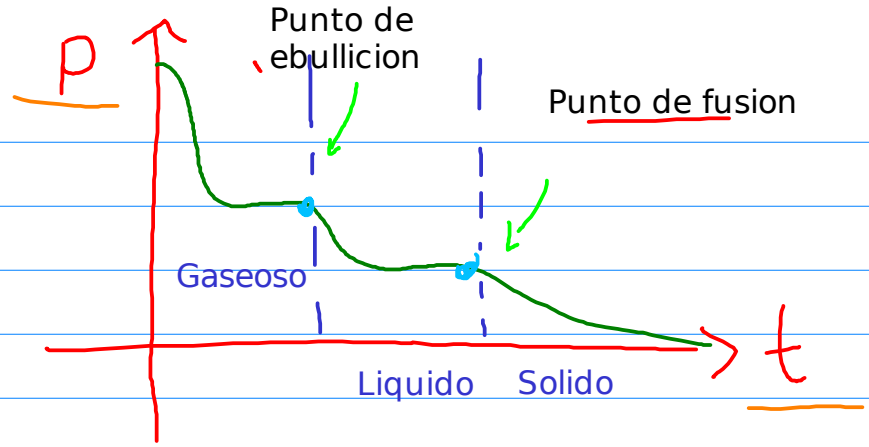
presion → pascales (Pa)

volumen → m3, cm3, mm3

temperatura → °C, K

} Propiedades cuantitativa

Propiedades intensivas



Solubilidad → liquidos con solidos

Concentracion → densidad

$m/v = \rho$

Propiedades extensivas

masa temperatura volumen

longitud

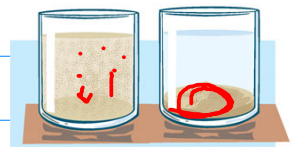
Solubilidad

Menos soluble Mas soluble

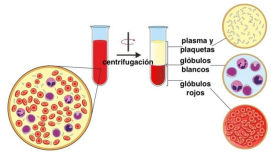
Heterogenea Homogeneo

Metodos de separacion de mezclas

sedementacion

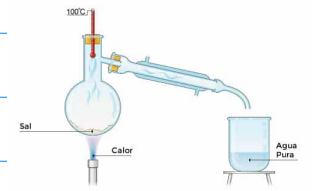


centrifugación



evapacion:

destilacion



cciones
s.

Mezclas

→ solido-solido
solido-gas
→ gas-gas

Disoluciones acuosas

→ solido-liquido
gas-liquido
liquido-liquido

Puras

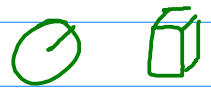
O_2
O_3
H_2

Compuestos

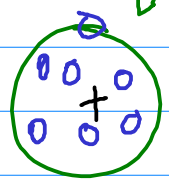
CO
NaCl

Elementos

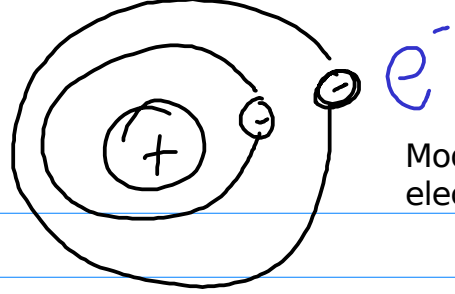
C
N
Li



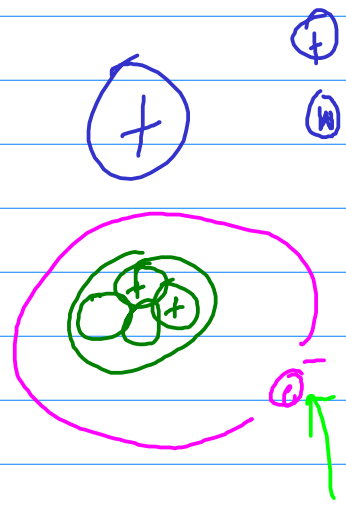
Atomo



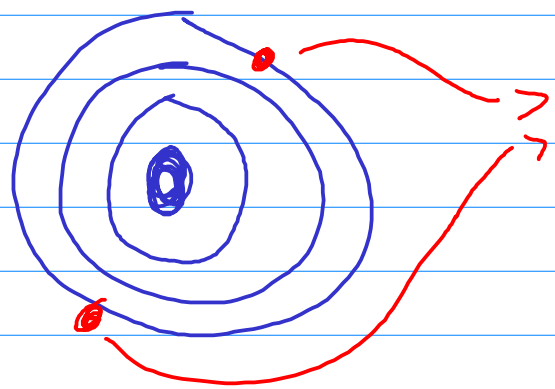
Modelo de Thomson (de pan con pasas)



Modelo de Rutherford (descubrimiento de electron)

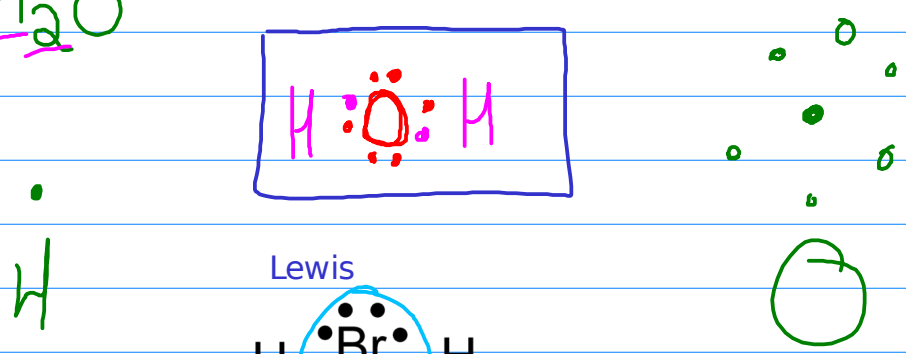


Modelo de Bohr

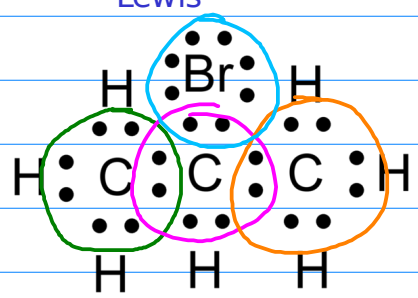


Electrones de valencia

H_2O



Lewis



Hidrógeno	H•	H•
Carbono	•C•	•C•
Agua	H:O:H	H-O-H
Etileno	H H C::C H H	H H C=C H H
Acetileno	H:C:::C:H	H-C≡C-H

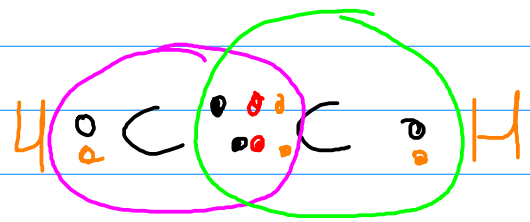
C

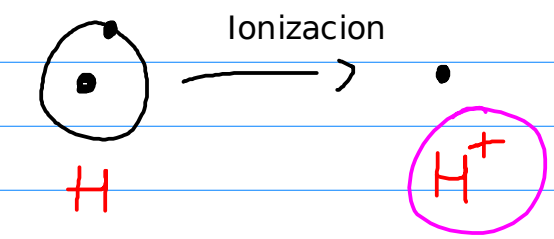
H

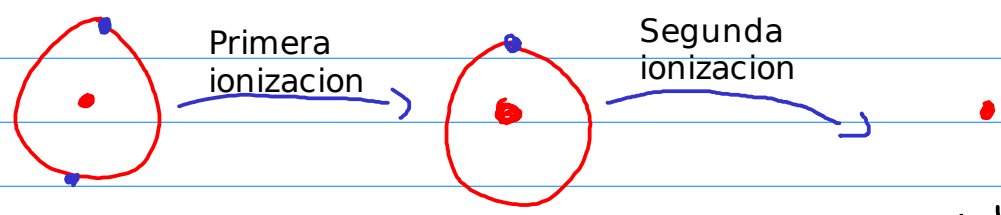
$\boxed{C_2H_2}$

C – 4

H – 1







He $\quad He^+ \quad He^{+2}$

Metales | Semimetales | No metales

3-5 4-4

Covalente

Se produce cuando dos elementos se "unen" para alcanzar el octeto, para que esten en sistema estable.

Ionico

Se producen en elementos clasificados como metales para alcanzar el octeto junto con elementos que tienen pocos electrones de valencia

$OH_2 - H_2O$

Puente de hidrogeno

H - H

Fuerzas de Van-der-Waals